



SRBS INTERNATIONAL SCHOOL

School Bus Tracking App

Official Product Documentation

VERSION 1.0

Nauhjheel, Mathura (Uttar Pradesh), India

"The Fear of the Lords is the Beginning of Wisdom"

Knowledge is Everything.....

1. Introduction

The **SRBS International School App** is a secure, private Android application built exclusively for the students, parents, class teachers, bus drivers and administrators of SRBS International School, Nauhjheel, Mathura. Its core purpose: **no child should ever miss the school bus**. Parents see the bus LIVE on a map, get its speed and expected arrival time, receive alerts, view attendance, fees and school records — all in one app, in **English or Hindi**.

Property	Detail
App Name	SRBS International School
Version	1.0 (Android)
Platform	Android 6.0+ (arm64), APK size ~28 MB
Technology	Flutter (Dart) + Google Firebase (Auth, Firestore, Realtime DB)
Maps	OpenStreetMap (free, no API key)
Languages	English + Hindi (instant switch, chosen at first login)
Users	4 roles: Student/Parent, Class Teacher, Driver, Admin
Cost to school	Rs. 0 running cost (Firebase free plan, free maps)





2. Student / Parent Panel — Every Feature

2.1 Login & Security

Feature	Description
Simple login	No email needed — login with Class + Section + Roll Number + Password.
First-login flow	Step 1: choose language (English/Hindi). Step 2: must set own private password (school's temporary password works only once).
One-time login	Login once, stay logged in — even offline or on slow internet. Logout only if: user logs out, admin force-logout, account disabled, or login on another phone.
Single-device lock	One account works on only ONE phone at a time. New phone login automatically logs out the old phone.
Forgot password	No risky email links. School office/teacher resets it; new password can be sent to the parent on WhatsApp with one tap.
Screenshot block	Screenshots and screen recording are blocked in the whole app (student data cannot leak).

2.2 Live Bus Tracking (Map)

Feature	Description
Live map	Bus moves live on OpenStreetMap. Light and Dark map styles (choice is remembered).
Honest tracking	Bus appears ONLY during an active trip with a fresh GPS fix. Old/stale locations are never shown as live. Trip off = friendly waiting screen ("map opens automatically").
Blue guide line	A dotted blue line connects the bus to the student's stop with an ETA bubble ("bus 5 min") in the middle.
Live stats card	Bottom card shows: bus speed (km/h), reach time (ETA), distance to my stop, and last-update time with LIVE / WEAK SIGNAL badge.
Route display	Real recorded road path (learned automatically from the driver's first trip) with all stops; my stop is highlighted with a star.
Map controls	Zoom +/- buttons, follow-bus lock, tap-to-recenter.
ETA & schedule	Arrival estimates per stop; never over-promises — shows "approx" and hides ETA when data is stale.

2.3 Home Screen (Customisable)

Feature	Description
My Bus card	Bus number, plate, ON TRIP/IDLE status, live-tracking indicator.
Customise Home	Student chooses WHICH widgets appear and in WHAT ORDER (drag & drop) — like a personal dashboard.
10 widgets	Bus card, Track banner, Quick actions, Timetable, Fees, Next stop + progress, "I'm on the bus" check-in, Driver card, School notices, Recent alerts.
Route progress bar	Dots for each stop, bus emoji at current position, my stop starred, covered part turns green.
SOS button	One tap → confirm → instant phone call to school office in an emergency.
Greetings	Time-based greeting (Good Morning/Afternoon/Evening) with student's name and photo.



2.4 Profile, Attendance, Fees & More

Feature	Description
Digital ID card	Plastic-card style student ID with photo, class, roll, bus, blood group badge — usable as identity in school.
Profile photo	Student can add own photo from gallery (compressed, stored securely).
Personal details	Father/Mother name, DOB (dd-mm-yyyy Indian format), blood group, admission no., address, email, parent phone — always visible; missing items show "—".
Documents tracker	All 6 school documents (Birth Certificate, Aadhaar, TC, Photos, Marksheet, Address Proof) with Submitted / Pending status.
Attendance calendar	Full month calendar — green = present, red = absent, grey = no record. Present/Absent/Percent counters. Browse previous months.
Leave application	Send absence application from the app (from-date, to-date, reason up to 1000 characters with a ready example letter shown in the box). Track status: PENDING → APPROVED/REJECTED.
Fees	Total / Paid / Due with progress bar, due date, payment history (PhonePe style), and Pay-via-UPI button (opens GPay/PhonePe/Paytm with amount pre-filled).
Self-update	Student can update parent phone and address themselves (school records stay protected).
Notifications	In-app alerts: trip started, bus approaching, reached stop, reached school, delays, school announcements. Unread badge count.
Themes	4 app themes — Classic Blue, SRBS Sky (school colours), Fresh Green, Royal Navy — entire app recolours instantly.
Language	English <-> Hindi toggle in settings; every screen changes instantly, no restart.
Phone widget	Home-screen widget (like Duolingo) with school logo and time-based message; tap opens the app.



3. Class Teacher Panel

Feature	Description
Separate panel	Teachers get their own simple 4-tab app (My Class, Attendance, Applications, Profile) — not the admin panel.
My Class	Full list of own class students with photos, details and fees view.
Class-locked	A teacher can see and manage ONLY their own class — other classes are invisible (enforced by server rules).
Digital attendance	Tap to mark present/absent, "All Present" one-tap, save whole class in one go. Students see it instantly in their calendar.
Add students	Teacher can create student accounts for own class (class/section auto-locked to theirs).
Leave approvals	See class leave applications, approve/reject with one tap; student sees status immediately.
WhatsApp send	After creating an account or resetting a password — one tap opens WhatsApp with the login details message ready-typed for the parent's number.
Password reset	Reset any own-class student's password (full profile data is preserved).

4. Driver Panel

Feature	Description
One-button trip	Big START TRIP / END TRIP button — nothing complicated for the driver.
Auto GPS sharing	While the trip is on, the phone shares bus location every few seconds — works with screen off (foreground service notification).
Trip direction	Morning pickup or afternoon drop selection.
Route recording	The driver's first trip automatically records the real road path; students then see the true route on their maps.
My route view	Driver sees own stops in order with a map.
Privacy	GPS is used ONLY during an active trip. Trip ended = tracking stops completely.



5. Admin Panel (School Office)

Feature	Description
Dashboard	Live counts (students, drivers, buses, active trips) + recent activity feed.
Student management	Add/edit/disable/delete students; assign bus & stop; set full profile (photo, parents, DOB, blood group, documents, admission no., email); LOGGED-IN status pill for every student.
Teacher management	Create class-teacher accounts tied to a class-section.
Driver & bus management	Create drivers, buses (number, plate, capacity), assign drivers to buses.
Routes & stops	Create routes, order stops, set scheduled times, school location.
Live monitoring	All buses on one map with name tags; per-bus card: direction, speed, last update, and current location ("At Ram Nagar stop" / "Near X, 1.2 km").
Fees control	Set total/due-date, record payments (auto history), UPI ID for parent payments.
Announcements	Send notices/alerts to all students or specific buses.
Error Reports center	If ANY user's app has an error, an automatic report reaches the admin (what error, which screen, which user, what time, app version) — no screenshots needed. "Mark fixed" workflow.
Force logout	Kick any account off its device instantly (lost phone, misuse).
Password reset	Reset any student password — new account keeps FULL profile (photo, fees, documents, attendance) + WhatsApp send option.
Activity logs	Every admin action is recorded with time (audit trail).
Trip history	Past trips with times and stop coverage.



6. Powerful Points (Strengths)

#	Strength	Why it matters
1	Rs. 0 running cost	Free Firebase plan + free OpenStreetMap. No monthly bills, no API keys, no card needed.
2	Honest live tracking	Never shows a fake/old bus position. Parents can trust what they see.
3	Bank-grade security model	Server-side Firebase Security Rules on every single read/write. Roles enforced on the server, not just in the app.
4	No email needed for kids	Class+Roll+Password login — perfect for young students.
5	Single-device + forced password change	Stolen passwords are useless: one device at a time, and the school's temporary password must be changed at first login.
6	Screenshot/recording blocked	Student photos, fees and location cannot be captured or leaked from the app.
7	Bilingual (English/Hindi)	Every screen, instantly switchable — parents in villages and cities both comfortable.
8	Smart error reporting	Problems auto-report to the admin with full details. The school knows about issues before users complain.
9	WhatsApp credential delivery	Login details reach parents on WhatsApp in one tap — no paper slips.
10	Self-learning routes	The app learns the real road path from the driver's first trip automatically.
11	Complete school suite	Tracking + attendance + fees + leave applications + documents + notices — one app instead of five.
12	Personalisable	Students customise home screen order, theme colour and language.



7. Limitations (Honest & Transparent)

Every product has limits. These are the known ones, with reasons and workarounds:

#	Limitation	Reason	Workaround
1	No push notifications when app is fully closed	Free Firebase plan does not include Cloud Functions needed for server push.	Alerts appear whenever the app is opened; keep app in background for live alerts.
2	Bus visible only during driver's trip	By design (privacy + accuracy) — and driver's phone must have internet + GPS on.	Driver keeps mobile data on; school SIM recommended.
3	ETA is approximate	Traffic, weather and stops cannot be perfectly predicted from speed alone.	App always labels ETA as approximate and hides it when signal is weak.
4	~100 simultaneous live-map viewers	Free Realtime Database allows 100 concurrent connections.	Fine for ~500-700 students in practice; upgradeable later without code change.
5	Android only (no iPhone app yet)	iOS needs a Mac + Apple fee (USD 99/yr).	Code is iOS-ready; can be published later if school decides.
6	WhatsApp send needs one tap	Fully automatic sending requires paid WhatsApp Business API.	One tap = message ready-typed; admin just presses Send.
7	Old auth accounts remain after password reset	Free plan cannot delete auth users from the app; old account is disabled instead.	Disabled accounts cannot log in; harmless.
8	Internet required	Live tracking is real-time data.	Profile/attendance stay visible from cache offline; map needs data.
9	Screenshots blocked also for demos	Security feature applies to everyone.	A special demo build can be made for videos/marketing.



8. Conditions of Use

#	Condition
1	Accounts are created ONLY by the school (admin/class teacher). There is no public sign-up.
2	The app is for SRBS International School members only — students, parents, teachers, drivers, staff.
3	Students/parents must keep their password private. First-login password change is compulsory.
4	One account = one phone at a time. Logging in on a new phone logs out the old one.
5	Driver must start the trip in the app for tracking to work, and end it after the trip.
6	Bus/stop assignments, fees and school records are managed only by the school office.
7	When a student leaves the school, the admin permanently deletes the account and its data.
8	Misuse (sharing passwords, false SOS calls, fake leave applications) may lead to account deactivation by the school.
9	UPI fee payments are made directly to the school's UPI ID; the app only opens the payment app — it never touches the money.
10	The school should keep the Firebase account credentials and app signing key safe for future updates.

9. Privacy & Data Safety (Summary)

Point	Detail
What is tracked	ONLY the school bus (driver's phone during trips). Student phones are NEVER tracked.
Data storage	School's own Google Firebase project (industry standard, encrypted in transit).
No selling / no ads	Data is never sold or shared; the app has zero advertisements.
Who sees what	Student: own data. Teacher: own class only. Admin: all (for management). Students cannot see each other's personal data.
Children's safety	No self sign-up; used under school & parent supervision; parents can request data view/deletion via school office.
Full policy	Available inside the app (Settings → Privacy Policy, English + Hindi) and as a web page for Play Store.





10. Capacity & Performance

Item	Free-plan limit	Practical meaning
Total accounts	Unlimited	Any number of students/staff.
Comfortable scale	~500-700 students + 30 staff	Daily reads/writes stay well inside free quotas.
Live-map viewers	100 at the same second	Normal usage never hits this; users check at different times.
Database storage	1 GB	Roughly 5,000+ student profiles incl. compressed photos.
Location updates	Every few seconds per bus	Smooth movement on the map.
Upgrade path	Blaze plan (pay-as-you-go)	Even then, this app's usage typically stays within the free allowance (~Rs. 0/month).

10.1 Technology Details

Component	Technology
App framework	Flutter 3 (Dart) — single codebase, native performance, iOS-ready.
Login system	Firebase Authentication (email-password engine used with roll-number IDs).
Database	Cloud Firestore (profiles, attendance, fees, applications, notices, error reports, audit logs).
Live location	Firebase Realtime Database, Singapore region (low latency for India).
Maps	flutter_map + OpenStreetMap tiles (light/dark).
Security	Server-side security rules for every collection; role checks on the server; FLAG_SECURE screen protection.
Quality	0 static-analysis issues; 15/15 automated tests passing; release build.

11. Version & Support

Item	Detail
Current version	v1.0 — first official release (September 2026).
Distribution	Direct APK (WhatsApp/website) today; Google Play Store upload ready (privacy policy, target SDK, 64-bit — all requirements met).
Updates	New APK installs over the old one; user data is never lost.
Issue handling	Errors auto-report to the Admin -> Error Reports screen; school contacts the developer with report details.
Requirements	Android 6.0 or newer, ~60 MB free space, internet connection for live features.

— End of Document —

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